

Terraform - Installation & Setup

Terraform: concept, Quality Engineering use cases, first steps and official reference.

Goal of this manual

This manual guides you through the complete installation and configuration process, from preparing the environment to final validation. Follow the steps in order and use each verification checkpoint to confirm that the setup was completed correctly.

Before you start

Confirm the supported operating system, runtime/dependencies and version requirements in the current official documentation before installing. The technical checklist below mirrors the validated tool-specific procedure.

Where to run commands

On Windows use PowerShell or Command Prompt; on macOS/Linux use Terminal. After changing PATH or environment variables, close and reopen the terminal before validating again.

Official references

https: //developer.hashicorp.com/terraform/install

Step-by-step installation and setup

1. Follow this tool-specific step: Instale Terraform pelo método oficial da HashiCorp para seu SO.
2. Follow this tool-specific step: Abra novo terminal e execute terraform -version.
3. Follow this tool-specific step: Crie terraform-lab e um main.tf mínimo com terraform { required_version = ">= 1.0" }.
4. Follow this tool-specific step: Execute terraform fmt.
5. Follow this tool-specific step: Execute terraform init.
6. Follow this tool-specific step: Execute terraform validate.
7. Follow this tool-specific step: Só use terraform plan/apply quando tiver provider/credenciais e autorização em ambiente de teste.

Commands and validation

```
terraform -version
```

```
terraform fmt
```

```
terraform init
```

```
terraform validate
```

First working use

Complete the first working use described in the steps above and confirm the expected tool-specific result before adding integrations or advanced configuration.

Common problems and fixes

terraform não reconhecido: revise instalação/PATH.

init falha ao baixar provider: investigue rede/proxy/certificados.

Credencial inválida no plan/apply: é configuração do provider, não instalação do Terraform.

Completion criteria

Treat setup as complete when the version/execution checks work and the first use in this manual finishes without errors. Only then move to integrations or advanced configuration.